Third semester B. Tech (ECE) Mid-term test, October 2020

Signals and Systems (ECE-2155), Max. Mark: 20

Note: Answer all questions. Write the answer in sheet of paper. **Clearly write your name, Roll number, Register number and put your signature on your answer sheet.** Upload the scanned/photo copy (pdf document) of the answer within the scheduled time.

1. A discrete time signal is given as x[n] = |n| for |n| less than 4, and x[n] = 0 for all other values of n. Plot the following signals neatly. **4M**

   (i) x[n]u[1-n]

   (ii) x[n]u[1-2n]

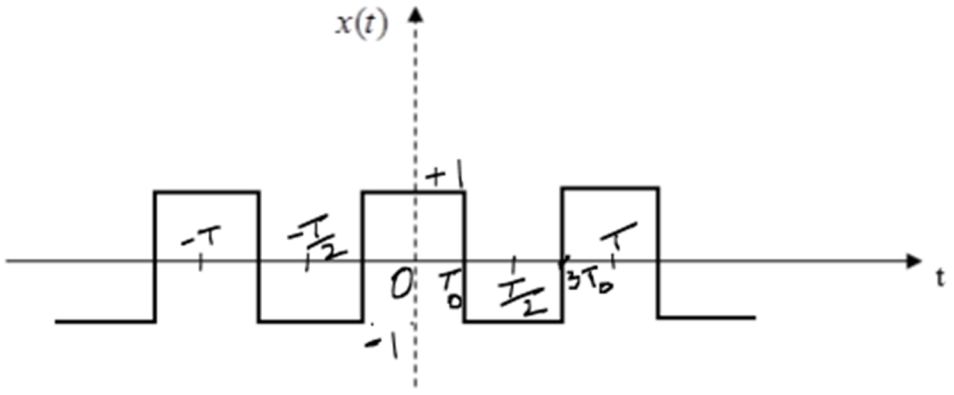
   (iii) x[n](u[n+2]-u[n])

   (iv) x[3n-2] u[-n+3]

2. Consider a continuous-time LTI system having the impulse response, $h(t) = e^{-2t}u(t-3)$ Using convolution, determine the response of the system for the input $x(t) = u(t+2) + u(t+1) - u(t-1) - u(t-2)$. **4M**

3. Determine the step response of LTI system whose impulse response is $h[n] = (-1)^n \{u[n+4] - u[n-5]\}$ **3M**

4. Consider the periodic signal x(t) given below.



   i) Determine the Fourier series coefficients for the signal and hence express the signal as sum of harmonically related cosines.

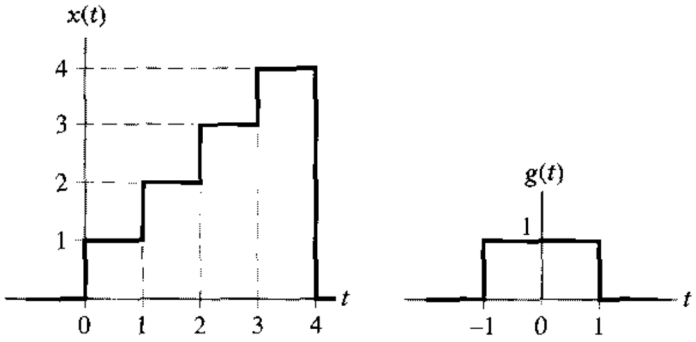
ii) This signal x(t) is input to a LTI system with frequency response

$$H(j\omega) = 1; 75\pi \leq |\omega| \leq 175\pi$$

$$= 0 \text{ elsewhere.}$$

Determine the output assuming $T_0 = 0.01$ and $T = 0.04$ **3M**

5. Express the following signal x(t) in terms of the signal g(t) giving detailed step wise explanation. **2M**



6. Sketch the waveform for the following equation and find the energy or average power of the signal as applicable. **2M**

x(t) = r(t+15) - r(t+10) - 10u(t-5) + r(t-5) - r(t-10)

7. Determine the time domain signal corresponding to the DTFT function $X(e^{j\Omega}) = j\sin(\Omega)$. Plot the signal. **2M**